

Components of boundary $\mathbb{Z}_5$ -flows on $K_{2,r}$

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Abstract

We classify the connected components of the nowhere-zero boundary-flow reconfiguration graph on $K_{2,r}$ over $\mathbb{Z}_5$, for every $r \geq 3$ and every boundary with total sum zero. Two flows are adjacent when the support of their difference is one cycle. Every boundary admits a flow. If the middle boundary values do not all lie in one nonzero pair $\{c, -c\}$, the graph is connected, and its order is given by a five-residue convolution. Otherwise, after normalization, its components are exactly the integer-sum levels $k \equiv q \pmod{5}$ in $\{0, 1, 2\}^r$, with sizes $[X^k](1 + X + X^2)^r$. In particular, exactly $2^{r+1} \min\{5, 2r - 4\}$ labeled boundaries are disconnected. The proof combines an integer-level invariant with a connectivity extension from three coordinates; its only finite base consists of five explicitly displayed spanning trees on 27 states.

1 The classification

Identify $\mathbb{Z}_5$ with the field $\mathbb{F}_5$. Let $H_r = K_{2,r}$ have vertices $u, v, w_1, \dots, w_r$, and orient its edges as $u \rightarrow w_i$ and $v \rightarrow w_i$. A boundary is a function $\beta : V(H_r) \rightarrow \mathbb{F}_5$ satisfying $\sum_x \beta(x) = 0$. A nowhere-zero β -flow assigns a value in $\mathbb{F}_5 \setminus \{0\}$ to each edge so that, at every vertex, the sum on outgoing edges minus the sum on incoming edges equals β . The vertices of $\mathcal{F}(H_r, \beta)$ are these flows. Distinct flows f, g are adjacent precisely when $\text{supp}(f - g)$ is a connected 2-regular subgraph of H_r . Thus the support must be a single 4-cycle, not a union of cycles.

Write $b_i = \beta(w_i)$, $s = \beta(u)$, and

$$D_b = \mathbb{F}_5 \setminus \{0, -b\}.$$

The repeated forbidden value when $b = 0$ is removed only once, so $|D_0| = 4$ and $|D_b| = 3$ for $b \neq 0$.

Theorem 1. *For every integer $r \geq 3$ and every total-zero boundary β , the graph $\mathcal{F}(H_r, \beta)$ is nonempty and has the following components.*

1. *If there is no nonzero $c \in \mathbb{F}_5$ with $b_i \in \{c, -c\}$ for all i , the graph is connected. Its unique component has size $N_r(s)$, where, for $t \in \mathbb{F}_5$,*

$$N_0(t) = \begin{cases} 1 & t = 0, \\ 0 & t \neq 0, \end{cases} \quad N_j(t) = \sum_{a \in D_{b_j}} N_{j-1}(t - a). \quad (1)$$

2. *Otherwise, choose $c \in \{1, 2\}$ such that all $b_i \in \{c, -c\}$. Set $n_- = |\{i : b_i = -c\}|$, and let $q \in \{0, 1, 2, 3, 4\}$ satisfy*

$$q \equiv c^{-1}s - r - n_- \pmod{5}. \quad (2)$$

There is exactly one component for every integer $k \in [0, 2r]$ with $k \equiv q \pmod{5}$, and its size is

$$C_r(k) = [X^k](1 + X + X^2)^r. \quad (3)$$

Consequently the number of components is

$$1 + \left\lfloor \frac{2r - q}{5} \right\rfloor, \quad (4)$$

and the graph is disconnected if and only if $q + 5 \leq 2r$.

All these quantities depend only on s and the multiset of the b_i ; the value $\beta(v) = -s - \sum_i b_i$ is forced.

For direct computation, the coefficients in (3) satisfy

$$C_0(k) = \mathbf{1}_{k=0}, \quad C_j(k) = C_{j-1}(k) + C_{j-1}(k-1) + C_{j-1}(k-2), \quad (5)$$

with $C_j(k) = 0$ outside $0 \leq k \leq 2j$. Equivalently,

$$C_r(k) = \sum_{h=\max\{0, k-r\}}^{\lfloor k/2 \rfloor} \frac{r!}{h! (k-2h)! (r-k+h)!}. \quad (6)$$

These formulas compute every component size without enumerating the reconfiguration graph.

Context and scope. Esperet, Hendrey, Lagoutte, Marseloo, Norin, and Steiner introduce the flow reconfiguration framework and study its relation to planar recoloring [2]. Cranston, Li, Su, Wang, and Xu formulate flow reconfiguration contractibility (FRC); their Definition 5.8 includes connectivity for every boundary, Theorem 5.9 gives a contraction application, and Theorem 5.11 classifies FRC cycles [1]. A nonempty fixed-boundary flow graph of a cycle is a clique. The present classification describes all components for the several-path family $K_{2,r}$, where their number can grow without bound.

The planar-dual results in [2, Section 5], including the dipole result, supply context for the zero-boundary case. Section 8 of that work also discusses extensions to boundary flows using group coloring. Care is needed with parallel dual edges: they may be discarded for ordinary coloring, as in the passage before its Corollary 5.8, but distinct group-coloring constraints on parallel edges need not be redundant. Section 5 gives an explicit disconnected boundary on $K_{2,3}$; hence the simple-dual connectivity conclusion cannot be transferred verbatim to every boundary. This does not contradict the stated zero-boundary result.

As broad adjacent context for nowhere-zero vectors subject to linear conditions over finite fields, we also cite the work of Zeng, Liu, and Ratnavelu on the Alon–Jaeger–Tarsi problem [3]. No result from that work or from the flow papers is needed for our proof. The comparison here is limited to the specified versions and inspected passages of the two flow papers. We found no equivalent classification there; wider literature and later versions have not been checked, and we make no claim of global priority.

2 Coordinates and connectivity extension

Let $a_i = f(uw_i)$. Conservation at w_i forces $f(vw_i) = -b_i - a_i$. The flow graph is therefore isomorphic to the graph on

$$\left\{ (a_1, \dots, a_r) \in \prod_{i=1}^r D_{b_i} : \sum_{i=1}^r a_i = s \right\}, \quad (7)$$

where two distinct tuples are adjacent exactly when two coordinates differ. Indeed, changing a_i changes both edges of the corresponding path, and a 4-cycle consists of exactly two such paths. An update thus takes a_i to $a_i + t$ and a_j to $a_j - t$, with $t \neq 0$, whenever the new entries remain in their respective domains. Conservation at v follows from the total-zero boundary condition.

We will repeatedly use the elementary fact

$$|A|, |B| \geq 3, \quad A, B \subseteq \mathbb{F}_5 \quad \implies \quad A + B = \mathbb{F}_5. \quad (8)$$

For every $t \in \mathbb{F}_5$, the sets A and $t - B$ intersect, since their sizes sum to more than five. Fixing all but two coordinates in (7) and applying (8) proves nonemptiness for every sum, already when $r \geq 2$. In particular, no empty-graph convention affects Theorem 1.

For nonempty domains $D_1, \dots, D_m \subseteq \mathbb{F}_5$, denote by $G(D_1, \dots, D_m; t)$ their sum- t fiber graph, with the same two-coordinate adjacency.

Lemma 2 (Connectivity extension). *Suppose $m \geq 3$, each $|D_i| \geq 3$, and $G(D_1, \dots, D_m; t)$ is connected for every $t \in \mathbb{F}_5$. For every nonempty $E \subseteq \mathbb{F}_5$ and every $s \in \mathbb{F}_5$, the graph $G(D_1, \dots, D_m, E; s)$ is connected.*

Proof. Slice the graph by its last coordinate $z \in E$. Each slice is the nonempty connected graph $G(D_1, \dots, D_m; s - z)$. For distinct z, z' , put $d = z' - z$. Choose $x \in D_1 \cap (D_1 + d)$; this intersection is nonempty because $2|D_1| > 5$, and $x - d \in D_1$. The remaining $m - 1 \geq 2$ domains can realize every sum by (8). Choose $x_2, \dots, x_m$ with total $s - z - x$. There is then an edge

$$(x, x_2, \dots, x_m, z) \sim (x - d, x_2, \dots, x_m, z').$$

Thus every pair of slices is joined by an edge. Their internal connectivity proves the claim. $\square$

The assumption about all five old sums is essential to this argument. It provides connectivity within each slice before a bridge is used; it does not assert that every state can immediately change its last coordinate to any prescribed value.

3 The three-coordinate base

Lemma 3. *Suppose no nonzero sign pair contains b_1, b_2, b_3 . Then the graph $G(D_{b_1}, D_{b_2}, D_{b_3}; s)$ is connected for every $s \in \mathbb{F}_5$.*

Proof. First suppose some $b_i = 0$, and permute coordinates so that $|D_{b_1}| = 4$. Slice by $z \in D_{b_3}$. Every two-coordinate sum fiber is a nonempty clique: two distinct pairs with the same sum differ in both coordinates. For distinct z, z' , put $d = z' - z$. The set $D_{b_1} \cap (D_{b_1} + d)$ has size at least $4 + 4 - 5 = 3$, and hence intersects $s - z - D_{b_2}$, which also has size at least three. For x in this intersection, put $y = s - z - x$. The edge $(x, y, z) \sim (x - d, y, z')$ joins the two slices. This proves the first case.

Otherwise all three b_i are nonzero and both sign pairs $\{1, 4\}$ and $\{2, 3\}$ occur. The identity

$$D_b + b = D_{-b} \quad (9)$$

allows a sign change of a single middle boundary: translate its coordinate by b and replace b by $-b$. This is an isomorphism, with the total sum changed from s to $s + b$. Coordinate permutations and simultaneous multiplication of coordinates, boundaries, and sum by a nonzero scalar are also isomorphisms.

Two of the three boundaries belong to one sign pair. Use (9) to make them both c , then scale by c^{-1} to make them both 1. The remaining boundary is 2 or 3, so a further sign change if necessary gives the prototype $(1, 1, 2)$. All transformations may change the total sum, which is why every s is checked below.

For this prototype the domains are $\{1, 2, 3\}, \{1, 2, 3\}, \{1, 2, 4\}$. Table 1 lists every state and a spanning tree in each fiber. The five lists contain 27 distinct allowed tuples, have the indicated sums modulo five, and therefore exhaust the 3^3 tuples. Each tree edge changes exactly two coordinates. This proves connectivity of every prototype fiber and completes the reduction. $\square$

s	Ordered states	Tree edges
0	122, 131, 212, 221, 311, 334	01, 02, 03, 14, 15
1	114, 132, 222, 231, 312, 321	01, 04, 12, 13, 45
2	124, 214, 232, 322, 331	01, 03, 12, 34
3	111, 134, 224, 314, 332	01, 03, 12, 14
4	112, 121, 211, 234, 324	01, 02, 14, 23

Table 1: The five fibers for $(b_1, b_2, b_3) = (1, 1, 2)$. The notation abc means (a, b, c) , not an integer. States are indexed from zero within each row; ij in the last column is the edge joining indices i and j .

Proof of Theorem 1, case 1. If some $b_i = 0$, select that coordinate and any other two. Otherwise, select one coordinate from each sign pair and any third coordinate. The selected triple satisfies Lemma 3. Reorder it first and append the remaining coordinates one at a time using Lemma 2. Every appended domain has size at least three, so the extension may be repeated. This proves connectivity for every $r \geq 3$.

The recurrence (1) counts allowed tuples according to their sum. Its successive convolutions commute, so the resulting order depends only on the multiset of the middle boundary values. $\square$

4 Components inside a sign pair

Proof of Theorem 1, case 2. Multiply all flow and boundary values by c^{-1} . The first edge at coordinate i then has integer representatives $\{1, 2, 3\}$ when $b_i/c = 1$, and $\{2, 3, 4\}$ when $b_i/c = -1$, where b_i denotes the original boundary value. Write

$$x_i = \ell_i + y_i, \quad \ell_i = \begin{cases} 1 & b_i/c = 1, \\ 2 & b_i/c = -1, \end{cases} \quad y_i \in \{0, 1, 2\}.$$

As $\sum_i \ell_i = r + n_-$, the state condition is $\sum_i y_i \equiv q \pmod{5}$.

Every legal move changes two coordinates x_i, x_j . Their possible integer pair sums lie in $[\ell_i + \ell_j, \ell_i + \ell_j + 4]$. Two integers in this interval that are congruent modulo five must be equal. Thus every move preserves the integer

$$k = \sum_{i=1}^r y_i. \tag{10}$$

There is no edge between distinct integer levels, even if their sums are congruent modulo five.

Conversely, let $y, z \in \{0, 1, 2\}^r$ have the same integer sum. If $y \neq z$, choose i with $y_i > z_i$ and j with $y_j < z_j$. Replacing (y_i, y_j) by $(y_i - 1, y_j + 1)$ stays in $\{0, 1, 2\}^2$ and is a legal two-coordinate move. It decreases $\sum_i |y_i - z_i|$ by two. Repeating reaches z , so each level is connected.

Every integer $k \in [0, 2r]$ occurs: use $\lfloor k/2 \rfloor$ entries equal to 2 and, if needed, one entry equal to 1. Hence the components are exactly the levels stated. Their sizes are $[X^k](1 + X + X^2)^r$. Choosing h entries equal to 2, $k - 2h$ equal to 1, and $r - k + h$ equal to 0 gives (6); adjoining one entry gives (5).

Since $q \leq 4 \leq 2r$, the admissible levels are $q, q + 5, \dots, q + 5\lfloor (2r - q)/5 \rfloor$. This proves (4) and the disconnection criterion. Replacing c by $-c$ reflects each y_i to $2 - y_i$ and k to $2r - k$, so the multiset of component sizes is independent of that choice. For fixed c , the formulas use only s, r , and n_- , completing the asserted dependence on boundary multiplicities. $\square$

Corollary 4. *Among the 5^{r+1} labeled total-zero boundaries on H_r , precisely*

$$2^{r+1} \min\{5, 2r - 4\}$$

give a disconnected flow graph.

Proof. There are two disjoint nonzero sign pairs, each allowing 2^r labeled middle boundary tuples. For each such tuple, s runs bijectively through $q = 0, \dots, 4$. The inequality $q \leq 2r - 5$ holds for exactly $\min\{5, 2r - 4\}$ of these residues. All other middle tuples are connected by Theorem 1. $\square$

5 Examples and exact checks

For $r = 3$ the coefficient list is

$$(C_3(0), \dots, C_3(6)) = (1, 3, 6, 7, 6, 3, 1).$$

For the boundary $(\beta(u), \beta(v), b_1, b_2, b_3) = (3, 4, 1, 1, 1)$, we have $q = 0$. The state $(1, 1, 1)$ is isolated, while the three permutations of $(2, 3, 3)$ form the other component. These are the levels $k = 0$ and $k = 5$ after subtracting 1 in each coordinate. This example explains the boundary-sensitive caution in the related-work discussion: the underlying simple plane dual of $K_{2,3}$ is a triangle, but its ordinary-coloring simplification does not retain every boundary constraint.

For $r = 4$,

$$(C_4(0), \dots, C_4(8)) = (1, 4, 10, 16, 19, 16, 10, 4, 1).$$

In particular, $q = 1$ gives component sizes $4 + 10$. Thus even states that are not isolated need not share a component. Table 2 records all disconnected size patterns for $r = 3, 4, 5$, counting labeled boundaries; all remaining boundaries are connected.

r	Connected boundaries	Disconnected sizes	Number of boundaries
3	593	$3 + 1$	32
4	2,997	$10 + 4$	64
		$16 + 1$	64
5	15,305	$51 + 1 + 1$	64
		$45 + 5$	128
		$30 + 15$	128

Table 2: Complete labeled-boundary counts in the first three values of r .

Exact computations reconstructed flows from all 4^{2r} nonzero edge assignments for $r = 3, 4, 5$, namely 4,096, 65,536, and 1,048,576 assignments. For $r = 3$ and $r = 5$, the component graphs were checked on all sorted-middle-boundary representatives, respectively 175 and 630 fibers. Their

labeled multiplicities are $r! / \prod_{b \in \mathbb{F}_5} n_b!$. For $r = 4$, all 3,125 labeled boundary graphs were rebuilt directly. The separate verifier determined adjacency from the connected degree-two support of the difference of the full edge assignments. For the $r = 3, 5$ certificates it checked complete state sets, adjacency lists, component labels, closure, and spanning trees. These checks recover Table 2.

The accompanying verification script also checks the literal prototype table and the component-size formulas. The finite enumeration supports the proof but is not its justification for unbounded r : that step is Lemma 2. The 27-state base in Table 1 is small enough to verify directly from the definitions. The original classification question is thereby resolved for precisely $\mathbb{Z}_5$, $K_{2,r}$, $r \geq 3$, and the single-cycle adjacency specified above; no statement for other groups or graphs is asserted.

Computational and AI assistance. Generative AI assisted with the coordinate reduction, proof development, exact enumeration code, and manuscript preparation. A separate model review reconstructed the argument and implemented the full-edge finite checks described above without importing the proposer’s code. This was a separate mathematical checking pass, not external peer review or statistically independent evidence. All reported computations used exact integer or finite-field arithmetic. No proof assistant was used; the self-contained mathematical argument and displayed finite base, rather than the software outputs, establish the theorem.

References

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